

JIEMING ZHANG

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EDUCATION

University of Virginia

Charlottesville, VA

B.S. Computer Engineering & B.S. Electrical Engineering

Expected Graduation: May 2029

Relevant Coursework: Computer Systems Organization, Electric Circuits, Digital Logic Design, CPU Architecture, Ordinary Differential Equations, Multivariable Calculus, Data Structures & C++

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Java

Hardware & EDA: Custom PCB Design, Power Management (3.7V LiPo to 3.3V LDO), KiCad, Altium, Quartus, ModelSim, ESP32, BNO055 IMU, MicroSD SPI, Oscilloscopes, Multimeters

Protocols & Tools: I2C, SPI, UART, CAN Bus, Fusion 360, Git, Matplotlib, SciPy

Concepts: Low-Level Firmware, Digital Logic, CPU Architecture, Signal Processing, Sensor Fusion

TECHNICAL PROJECTS

IoT Golf Swing Analyzer & Visualizer Prototype

Summer 2026

Embedded Hardware, Firmware & Python Analytics · Personal Project

- Designed and routed a custom PCB featuring power management circuits to convert a 3.7V LiPo battery input to a regulated 3.3V rail for an ESP32 microcontroller.
- Developed non-blocking C++ embedded firmware using ESP32 to interface with a BNO055 9-axis IMU over I2C and stream 100Hz 3D quaternion and accelerometer data to custom SPI MicroSD storage.
- Created a 3D Python telemetry engine using Matplotlib and SciPy to process raw quaternions, diminish gravitational drift, estimate clubhead velocity, and render dynamic swing trajectories.

CPU Datapath & Controller Design

Spring 2026

UVA ECE Coursework

- Architected a multi-component CPU datapath in structural VHDL, incorporating ALU functions, register banks, and control sequencers.
- Debugged timing signals and uninitialized control states in ModelSim by analyzing testbench reset pulses and waveform traces.
- Synthesized digital logic structures onto FPGA target devices using Intel Quartus.

Automated Physical Sink Shutoff System

Fall 2025

Introductory Engineering Design Project · UVA

- Engineered an external retrofitted automated shutoff mechanism for standard sink handles without internal plumbing access.
- Integrated an ultrasonic distance sensor to detect user presence and trigger automated shutoff sequences.
- Designed and calibrated a servo-driven mechanical actuation arm to physically manipulate the external water valve handle.

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE)

Fall 2025 – Present

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Design schematics and multi-layer PCBs using Altium and KiCad for sensor integration and data logging on an all-electric FSAE racecar.
- Build electrical wiring harnesses and integrate chassis hardware components to ensure noise-resistant telemetry transmission.
- Model custom mechanical mounting enclosures in Fusion 360 for DAQ subteam vehicle installations.

LEADERSHIP & EXPERIENCE

Mathnasium

Fall 2022 – Jan 2023

Math Tutor & Trainer

- Instructed up to 16 students per session across subjects ranging from arithmetic through AP Calculus.
- Onboarded and trained new instructional staff on curriculum delivery and student engagement strategies.